THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHS · HARD LEVEL CHAPTER TEST SET 3

Chapter 10: **Circles** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A circle has centre O. Tangents at A and B meet at P and $\angle AOB = 2\angle APB$. Then $\angle APB$ equals: 1

(a) 45° **(b)** 30° **(c)** 90° **(d)** 60°

2. Tangents PA and PB are drawn from a point P, 35 cm from the centre O of a circle of radius 21 cm. The length of each tangent is 28 cm. The length of the chord AB is: 1

(a) 33.6 cm **(b)** 16.8 cm **(c)** 56 cm **(d)** 42 cm

3. PA and PB are tangents from P to a circle, each of length 19 cm. A third tangent touches the circle at E on the minor arc AB and meets PA at C and PB at D. The perimeter of $\triangle PCD$ is: 1

(a) 19 cm **(b)** 21 cm **(c)** 57 cm **(d)** 38 cm

4. Tangents QA and QB are drawn from an external point Q to a circle, and $QA = (14x + 6)$ cm and $QB = (8x + 30)$ cm. The length of QA is: 1

(a) 64 cm **(b)** 86 cm **(c)** 4 cm **(d)** 62 cm

5. Two concentric circles have radii 74 cm and 24 cm. A chord of the larger circle touches the smaller circle. The length of the chord is: 1

(a) 70 cm **(b)** 98 cm **(c)** 140 cm **(d)** 100 cm

6. In $\triangle ABC$, right-angled at B, $AB = 18$ cm and $BC = 80$ cm. A circle is inscribed in the triangle. The radius of the inscribed circle is: 1

(a) 41 cm **(b)** 9 cm **(c)** 8 cm **(d)** 16 cm

7. A point P is 87 cm from the centre O of a circle of radius 63 cm. The length of the tangent drawn from P to the circle is: 1

(a) 60 cm **(b)** 150 cm **(c)** 3600 cm **(d)** 24 cm

8. Two tangents are drawn from P to a circle of radius r and the angle between them is 60°. The distance OP is: 1

(a) $\sqrt{3}\,r$ **(b)** $\sqrt{2}\,r$ **(c)** $2r$ **(d)** r

9. **Assertion (A):** A circle has radius 3 cm. The length of the tangent from a point P at distance 5 cm from the centre is 4 cm. 1

Reason (R): The tangent at a point is perpendicular to the radius through that point, so the Pythagoras theorem can be applied.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

10. **Assertion (A):** If PA and PB are tangents from an external point P to a circle with centre O, then $\angle AOB + \angle APB = 180°$. 1

Reason (R): The tangent at a point of a circle is parallel to the radius through that point.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Two concentric circles have radii 95 cm and 76 cm. A chord of the larger circle touches the smaller circle. Find the length of the chord. 2

12. Tangents PA and PB are drawn from a point P, 55 cm from the centre O of a circle of radius 44 cm. The length of each tangent is 33 cm. Find the length of the chord AB. 2

13. PA and PB are tangents from P to a circle, each of length 28 cm. A third tangent touches the circle at E on the minor arc AB and meets PA at C and PB at D. Find the perimeter of $\triangle PCD$. 2

14. Two concentric circles have radii 97 cm and 72 cm. A chord of the larger circle touches the smaller circle. Find the length of the chord. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Tangents PA and PB are drawn from a point P, 75 cm from the centre O of a circle of radius 21 cm. The length of each tangent is 72 cm. Find the length of the chord AB. 3

16. In $\triangle ABC$, right-angled at B, $AB = 39$ cm and $BC = 80$ cm. A circle is inscribed in the triangle. Find the radius of the inscribed circle. 3

17. Tangents QA and QB are drawn from an external point Q to a circle, and $QA = (20x + 8)$ cm and $QB = (12x + 24)$ cm. Find the length of QA. 3

OR

Tangents PA and PB are drawn from a point P, 45 cm from the centre O of a circle of radius 27 cm. The length of each tangent is 36 cm. Find the length of the chord AB. 3

18. In $\triangle ABC$, right-angled at B, $AB = 54$ cm and $BC = 72$ cm. A circle is inscribed in the triangle. Find the radius of the inscribed circle. 3

OR

Tangents QA and QB are drawn from an external point Q to a circle, and $QA = (25x + 3)$ cm and $QB = (17x + 19)$ cm. Find the length of QA. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** Two concentric circles have radii 13 cm and 5 cm. A chord AB of the larger circle touches the smaller circle at the point P. Centre is O. 5

(i) What is the measure of $\angle OPA$ and why?

(ii) Find the length AP.

(iii) Find the length of the chord AB.

(iv) If the radii were 17 cm and 8 cm instead, find the length of such a chord.

20. Tangents PA and PB are drawn from a point P, 75 cm from the centre O of a circle of radius 45 cm. The length of each tangent is 60 cm. Find the length of the chord AB. 5

OR

In $\triangle ABC$, $\angle B = 90°$, $AB = 65$ cm and $BC = 72$ cm. A circle touching all three sides touches AB at F and BC at D. Find the radius of the circle and the lengths AF and CD (in this order). 5

END OF QUESTION PAPER — **Total: 40 Marks**